

Literature Review

Predictive maintenance has become one of the most important applications of Artificial Intelligence and Machine Learning in industrial systems. Traditional maintenance approaches such as corrective maintenance and preventive maintenance often lead to high operational costs, unexpected machine failures, and production downtime. To overcome these challenges, researchers introduced predictive maintenance systems that monitor machine conditions continuously and predict failures before they occur.

Many studies have focused on using sensor data and audio signals to detect anomalies in industrial machines. Machine learning algorithms are capable of analyzing large amounts of operational data and identifying abnormal behavior patterns that may indicate machine degradation or future failure.

One of the widely used datasets in this field is the **MIMII Audio Dataset**, which contains normal and abnormal operating sounds collected from industrial machines such as fans, pumps, sliders, and valves. Researchers use this dataset to train deep learning models capable of distinguishing between healthy and faulty machine states through audio analysis.

In addition to audio-based monitoring, predictive maintenance has also been applied to turbofan engines using sensor-based datasets. A popular dataset used in this area is the **NASA Turbofan Engine Degradation Dataset**, which contains operational sensor measurements collected during engine life cycles. Machine learning models are trained on this data to estimate the Remaining Useful Life (RUL) of engines and predict possible failures before they happen.

Several machine learning and deep learning techniques have been proposed in previous studies, including:

- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN)
- Long Short-Term Memory (LSTM)
- Autoencoders
- Random Forest algorithms

These techniques achieved high accuracy in detecting anomalies and predicting failures. Deep learning models, especially CNN and LSTM, showed strong performance because they can automatically extract complex features from audio and sensor data.

Recent research also highlights the importance of feature extraction methods such as:

- Mel Spectrograms
- MFCC (Mel-Frequency Cepstral Coefficients)
- Time-series sensor analysis

These methods help improve the model's ability to identify hidden patterns in machine behavior.

The proposed project builds upon these previous studies by developing a predictive maintenance system that uses machine learning techniques to analyze machine data and detect anomalies. The system aims to improve maintenance efficiency, reduce downtime, and increase equipment reliability through intelligent failure prediction.